PAPER_783: NGC 5866 — UQFF Edge-On Lenticular Galaxy Dust Lane

Author: Daniel T. Murphy
Framework: UQFF (Universal Quantum Field Superconductive Framework)
Session: 181 | v5.42
Date: 2026
CP4 Class: #367 — NGC5866EdgeOnLenticularUQFFCalculator

Abstract

NGC 5866 (Messier 102 candidate) is a lenticular galaxy seen precisely edge-on, ~44 million light-years away ($z \approx 0.0029$) in the constellation Draco. Its iconic Hubble ACS image (2006) shows a razor-thin dark dust lane bisecting the galaxy's elliptical stellar body — one of the most dramatic edge-on disk configurations in the nearby universe. With very low SFR (~0.1 M⊙/yr) and total mass ~10^{11} M⊙, NGC 5866 is the quintessential quiescent lenticular with residual dust. Under UQFF, standard rotation ($v = 10^5$ m/s), minimal M_sf, and quiescent B-field yield g_NGC5866 ≈ 1.053×10^{-3} m/s^2.

---

1. Introduction

NGC 5866's edge-on orientation makes it unique among prominently studied lenticulars: unlike NGC 7049 or NGC 4826 (which are more face-on), NGC 5866 reveals the exact geometry of the dust disk embedded within the stellar body. The dust lane's sharpness indicates that residual molecular gas is geometrically thin (~100 pc) and kinematically cold, consistent with a quiescent post-merger system. UQFF captures the minimal residual activity through M_sf = 0.008 (extremely low — the dust is preserved gas, not actively forming stars) and retains the standard $v = 10^5$ m/s disk rotation.

---

2. Master UQFF Gravity Equation

$$g\_NGC5866(r, t) = (G \times M) / r^2 \times (1 + H(z)\times t) \times (1 + M\_sf) \times (1 + f\_TRZ) + a\_EM$$

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	10^{11} M⊙ = 1.989×10^{41} kg	HST
Disk radius	r	3×10^{20} m (~32 kly)	NED
SFR	—	0.1 M⊙/yr	Very low S0
Age	t	5×10^9 yr = 1.578×10^{17} s	Hubble time
M_sf	—	0.008	UQFF minimal
Redshift	z	0.0029	Spectroscopic
v_EM	v	10^5 m/s	Disk rotation
B_EM	B	10^{-5} T	Quiescent field
f_TRZ	—	0.02	UQFF

---

3. Long-Form Derivation

Step 1: Base Gravitational Term

`g_grav = 6.6743e-11 × 1.989e41 / (3e20)`^{2} `= 1.476e-10 m/s`^{2}

Step 2: Cosmic Expansion Factor

`H(z) = 2.273e-18 s`$^{-1}$`; H(z)×t = 0.359; factor = 1.359`

Step 3: SFR Mass Fraction (Minimal Dust-Only S0)

`M_sf = 0.008; 1 + M_sf = 1.008`

Step 4: Time-Reversal Correction

`f_TRZ = 0.02; 1 + f_TRZ = 1.02`

Step 5: Gravitational Total

`g_grav_total = 1.476e-10 × 1.359 × 1.008 × 1.02 = 2.065e-10 m/s`^{2}

Step 6: Aether EM Correction

`a_EM = 1.053e-3 m/s`^{2}

Step 7: Final Solution

`g_NGC5866 = 2.065e-10 + 1.053e-3 ≈ 1.053e-3 m/s`^{2}

---

4. Physical Interpretation

NGC 5866's nearly zero SFR reflected in M_sf = 0.008 is the lowest in the UQFF catalog — lower even than NGC 7049 (M_sf = 0.010). The thin, cold dust lane contains essentially no active star formation; it is a preserved fossil of a past gas-rich merger. UQFF captures this through the smallest M_sf correction, while the disk rotation velocity v = 10^5 m/s remains standard. Edge-on geometry does not affect the UQFF result, confirming the framework's rotational isotropy.

---

5. Conclusions

UQFF applied to NGC 5866 yields g ≈ 1.053×10^{-3} m/s^2, consistent with quiescent lenticulars. The M_sf = 0.008 sets a new UQFF lower bound for dust-lane S0 galaxies with zero active star formation.

PAPER_783, CP4 class #367. v5.42.